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Wave drag on human swimmers

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Abstract

Drag measurements from a towed mannequin show total drag at the surface is up to 2.4 times the drag when fully immersed. This additional drag is due to the energy required to form waves in the wake behind the mannequin. The measurements show that passive wave drag is the largest drag, comprising up to 50–60% of the total at $1.7\,\mathrm{m\,s^{-1}}$, much higher than any previous estimates. Comprehensive measurements spanning human swimming speeds and tow depths up to 1.0 m demonstrate that wave drag on the mannequin is less than 5% of total drag for tows deeper than 0.5 m at $1\,\mathrm{m\,s^{-1}}$ and 0.7 m at $2\,\mathrm{m\,s^{-1}}$. Wave drag sharply increases above these depths to a maximum of up to 60% of the mannequin's 100 N total drag when towed at the surface at $1.7\,\mathrm{m\,s^{-1}}$. The measurements show that to avoid significant wave drag during the underwater sections of starts and turns, swimmers must streamline at depths greater than 1.8 chest depths below the surface at Froude number (Fr) = 0.2, and 2.8 chest depths at $Fr = 0.42$. This corresponds to speeds of 0.9 and $2.0\,\mathrm{m\,s^{-1}}$, respectively, for a chest depth of 0.25 m and toe to finger length of 2.34 m.
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Keywords: Swimming; Wave; Drag; Measurement

1. Introduction

Swimmers and vessels travelling on the water's surface generate waves which form a wake behind them. When travelling at high speed, as measured by the Froude number (Fr), a large proportion of the drag on surface vessels is due to the energy required to create these waves (Van Manen and Van Oossanen, 1988). Elite swimmers also travel at high Fr numbers and the aim of this paper is to give the results of fundamental research which demonstrates that wave drag is the largest component of the drag they experience when at the surface.

There are three main types of drag on surface vessels; viscous or skin friction drag, form drag associated with the turbulent wake of the vessel and wave drag. Wave drag is due to the energy required to create the transverse and divergent waves which lie within a 39° sector behind the vessel. The drag is dependant on the ratio of its speed to that of a water wave with a wavelength equal to the vessel's length, i.e. the Froude number

$$Fr = \frac{V}{\sqrt{gL}},$$

where V = vessel speed, $g = 9.81\,\mathrm{m\,s^{-2}}$, L = length of the vessel. A typical total drag curve for a vessel is shown in Fig. 1. Above $Fr = 0.25$ the drag increases rapidly due to the increasing importance of wave drag. Drag curves may display "bumps" at particular speeds due to constructive and destructive interference of the waves generated at each change in cross-section of the hull. Around $Fr = 0.42$, the vessel's speed matches that of a wave which has a wavelength equal to the length of the vessel. This is the so-called "hull speed", the maximum that can be achieved as a displacement vessel. The drag increase is typically less rapid above $Fr = 0.45$, as the vessel generates hydrodynamic lift, reducing its displacement as it rises to a planing condition. A 1.8 m

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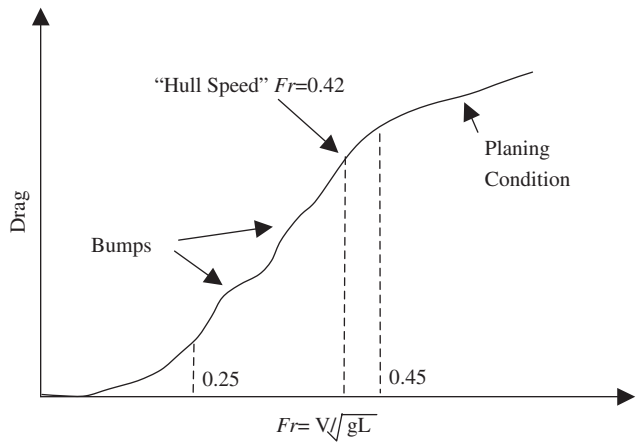

Fig. 1. Schematic drag curve for a surface vessel versus Froude number. Based on Van Manen and Van Oossanen (1988).

tall elite swimmer at $1.8\,\mathrm{m\,s^{-1}}$ has $Fr = 0.42$, and with arms extended to a total length of 2.3 m, has $Fr = 0.40$, thus wave drag would be expected to constitute a large proportion of their total drag.

Several attempts have been made to quantify wave drag on swimmers. Vorontsov and Rumyantsev (2000) suggested that 5% of drag was due to waves at $2\,\mathrm{m\,s^{-1}}$. Toussaint et al. (2002) used the Measurement of Active Drag (MAD) system measurements to estimate that wave drag had a greater contribution, 21% at $1.9\,\mathrm{m\,s^{-1}}$. Wilson and Thorp (2003) used power law fits to estimate that wave drag contributed between 10% and 20% of active drag at $1.0\,\mathrm{m\,s^{-1}}$ and between 35% and 45% at $2.0\,\mathrm{m\,s^{-1}}$. Lyttle et al. (1998) towed 40 swimmers using a pulley system at velocities from 1.6 to $3.1\,\mathrm{m\,s^{-1}}$ and depths between the surface and 0.6 m. They found that drag was 20% lower at 0.6 m depth than at the surface when towing at $2.2\,\mathrm{m\,s^{-1}}$.

A deeply immersed solid body towed horizontally does not generate surface waves and drag is mainly due to skin friction and form drag. Viewed from a perspective moving with the body, the effect of the body is to force fluid to flow around it, distorting the flow near the body. In some regions, the distortion increases velocities and in others reduces velocities relative to the undisturbed flow, with the size of the velocity distortions decreasing with distance from the body. When the body moves close enough to the water's surface for the distorted flow to impinge on the surface the pressure changes, due to the Bernoulli effect, associated with the distortion cause both depressions and elevations in the water's surface above the body. These in turn create a wave wake and thus the body begins to experience wave drag. The closer the body is to the surface, the larger the depressions and elevations, leading to greater wave drag. Wave drag is an additional drag experienced near the surface which is not present when deeply immersed, thus total drag would be expected to increase as the body approaches the surface. Wave drag is typically assumed independent of skin and form drag and thus the difference between the total drag measured when towed near to or at the surface and the drag when deeply immersed is an estimate of the drag due to waves.

2. Measurements

The measurements were made in a 2.5 m wide, 1.5 m deep flume with a 10 m long working section capable of water speeds up to $3\,\mathrm{m\,s^{-1}}$. The towing rig consisted of a forward strut joined to a horizontal towing rod in series with a load cell. To keep it horizontal, the rod passed through a frictionless sleeve in a rear strut (Fig. 2). The mannequin was attached to the rod 1 m behind the rear strut to reduce the effect of the flow around the struts on the mannequin. Underwater observations of the mannequin showed that the toe depth, which was not fixed, was very stable during tows. Tow depth was varied by vertically adjusting the horizontal towing rod. A Marsh McBinney Flow Mate 2000 velocity meter was attached on the rear strut just above the horizontal towing rod. The flume's velocity profiles both vertically and horizontally have been comprehensively measured and show variation of less than $0.1\,\mathrm{m\,s^{-1}}$ outside of the regions within 20 cm of the walls and floor. With no mannequin in the flow the water surface was flat within 2 cm at $2.6\,\mathrm{m\,s^{-1}}$, much less than the 15 cm wave generated by the mannequin at these speeds.

A Sensotec Model 31/4267-06 tension load cell was placed between the forward strut and the horizontal rod

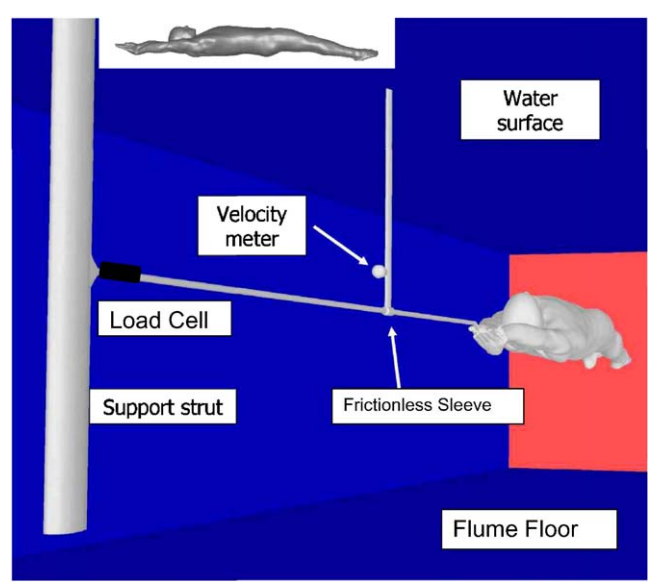

Fig. 2. Experimental setup of towing rig. A load cell was mounted between the horizontal towing rod and the front support strut. The male mannequin was towed face up with arms extended. Inset shows mannequin's streamlined posture.

to measure drag force on the mannequin. Measurements were recorded at 100 Hz using an AD Instruments Maclab 8e digitizer and were averaged for each 30 s tow run. The load cell was dry calibrated using weights and checked against a second calibrated load cell to an accuracy of 1 N.

Preliminary tow tests using a human subject showed that small changes in streamlined body posture altered drag by around 10% and the short duration of the tows meant limited averaging of forces, resulting in variable measurements. In addition, when towed near the surface human subjects found it difficult to consistently adopt the same streamlined body alignment for all tows. The variability associated with body posture was eliminated by using a fibreglass mannequin built using digital scans of an elite male swimmer. The mannequin was adjusted to have slight negative buoyancy, was 1.86 m tall, with a finger to toe length of 2.34 m, had head, chest, waist and hip circumferences of 0.59, 1.02, 0.8 and 0.98 m, respectively, a projected frontal area of 0.0934 m^2, a total surface area 1.8594 m^2 and a chest depth of 0.25 m.

The mannequin was towed at velocities from 0.4 to 2.6 $m\,s^{-1}$ at 0.2 $m\,s^{-1}$ intervals. The velocity meter was on the rear strut just above the horizontal rod and was not immersed for tow depths shallower than 0.3 m. The near surface tow velocities were taken as the mean velocity of all runs at the same flume drive settings. The mannequin was towed with its longitudinal body axis at depths from 1.0 to 0.0 m relative to the still water level, at 0.1 m depth intervals. The deepest tow was 0.5 m above the floor of the flume's working section. The reported tow depth is the depth of the longitudinal body axis at the hands. Separate runs without the mannequin were made at the same velocity intervals at depths of 0.0, 0.1, 0.3, 0.6 and 1.0 m and were used to adjust the 143 mannequin tows for the drag due to the rod.

3. Results

3.1. Drag curves

The drag–velocity curves in Fig. 3 show the increased drag with velocity. The curves from tow depths of 1.0–0.8 m are very similar, indicating weak interaction with the surface. A drag curve for the fully immersed mannequin was estimated by averaging the drag for towing depths 1.0–0.8 m at each velocity (the average is shown in Fig. 7). At 0.4 m depths and above drag rapidly increases, with drag at some velocities up to 2.4 times that when the mannequin was fully immersed at the same speed.

The five near surface drag–velocity curves exhibit distinct "bumps". For example, the 0.4–0.0 m curves show a high gradient between 1.1 and 1.9 $m\,s^{-1}$, and reduced gradient, or a drag decrease, around 2 $m\,s^{-1}$,

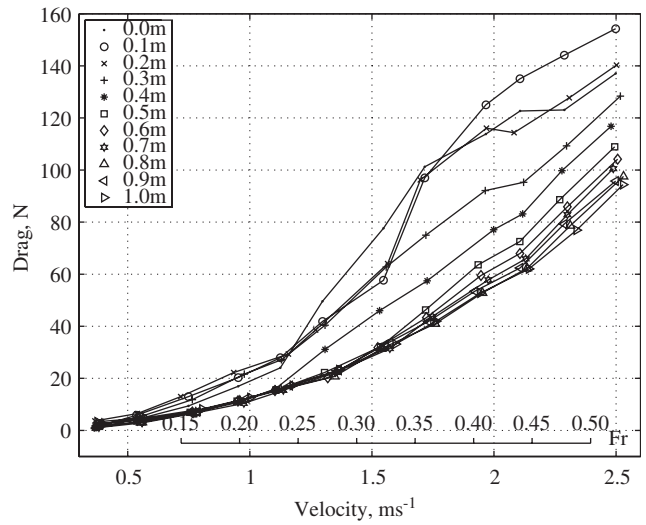

Fig. 3. Passive drag versus velocity for male mannequin at tow depths of 0.0–1.0 m. The Froude number, *Fr*, axis is based on finger to toe length.

giving a mid-range bump. Except for the 0.1 m curve, the change to a slower rate of increase in drag is at a distinct velocity near 2 $m\,s^{-1}$, suggesting an association with a change in flow patterns around the mannequin at this velocity. Observations of the flow showed that at speeds less than 1.0 $m\,s^{-1}$ and tow depths 0.3–0.0 m, the hips and head were above the surface. As speed increased the mannequin progressively immersed from the face down. The speed at which the mannequin was fully covered by water ranged from 2.6 $m\,s^{-1}$ at 0.0 m tow depth, to 1.8 $m\,s^{-1}$ at 0.3 m depth, which spans the velocities at the upper side of the drag–velocity bumps. Thus the mannequin may be more streamlined after its upper surface becomes fully covered by water at high speeds, but there was no clear correlation between the speeds at which this happens and the velocity at the upper side of the bumps. The bumps are more likely due to the progressive immersion, than to interference of waves produced by different parts of the mannequin.

Drag measurements are usually presented as dimensionless drag coefficients defined by

$$C_D = \frac{D}{1/2\rho A V^2},$$

where D is the measured drag, ρ the fluid density, "A" is typically the surface or frontal area. Fig. 4 shows the non-dimensional drag coefficient, based on frontal area, plotted against Fr based on finger to toe length. This plot shows that C_D when immersed increases for $Fr > 0.15$. The shallower tows show much higher drag coefficients and have more variability, due to the drag bumps. At the lowest speeds, where $Fr < 0.15$, the C_D estimates are probably not reliable as they result from the ratio of small numbers, where velocities are under 0.6 $m\,s^{-1}$ and drag values under 8 N.

3.2. Drag contours

Flow velocities and depths with the same measured drag are illustrated by the contour lines of drag which are plotted against velocity and depth in Fig. 5. The contours for depths between 1.0 and 0.8 m are near vertical, confirming that at these depths, there is no interaction of the distorted flow around the mannequin with the water's surface. For higher velocities and tow depths from 0.7 to 0.5 m, depth contours slope to the upper left, indicating that the same drag is achieved at a lower velocity, or conversely a higher drag occurs at the same velocity. Between 0.4 and 0.2 m the contour slope, with a decrease in depth, increases significantly, due to a large increase in drag at shallower tow depths. The details above 0.2 m are complex, with a drag maximum at 0.1 m at high velocities, $>1.7\,\mathrm{m\,s^{-1}}$, and the highest drags at 0.0 m at medium velocities, 1.2–1.7 $\mathrm{m\,s^{-1}}$. These features may result from the way in which the mannequin progressively immerses from head to hips with increasing velocity for tow depths less than 0.4 m.

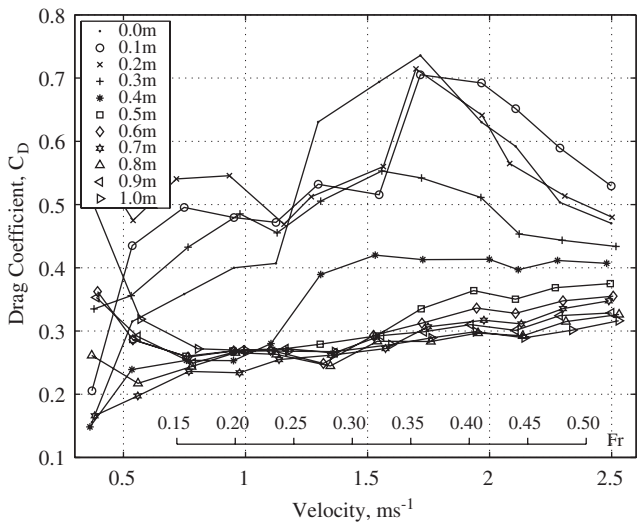

Fig. 4. Passive drag coefficient, based on projected frontal area, versus velocity and Froude number based on finger to toe length.

The changes in drag due to tow depth are clearly shown in Fig. 6 which gives contours of the additional drag above that of the fully immersed mannequin as a percentage of the total drag. Below 0.7 m the additional drag is near zero as the drag curves are similar to the fully immersed curve. Between 0.4 and 0.2 m at 1 $\mathrm{m\,s^{-1}}$ and between 0.5 and 0.2 m at 2 $\mathrm{m\,s^{-1}}$ in Fig. 6, drag increases rapidly. Closer to the surface the pattern is again more complex with a sub-surface maximum at 0.1 m above 1.7 $\mathrm{m\,s^{-1}}$.

3.3. Velocity power law

Drag curves are often presented as a power law relationship

$$D = aV^n,$$

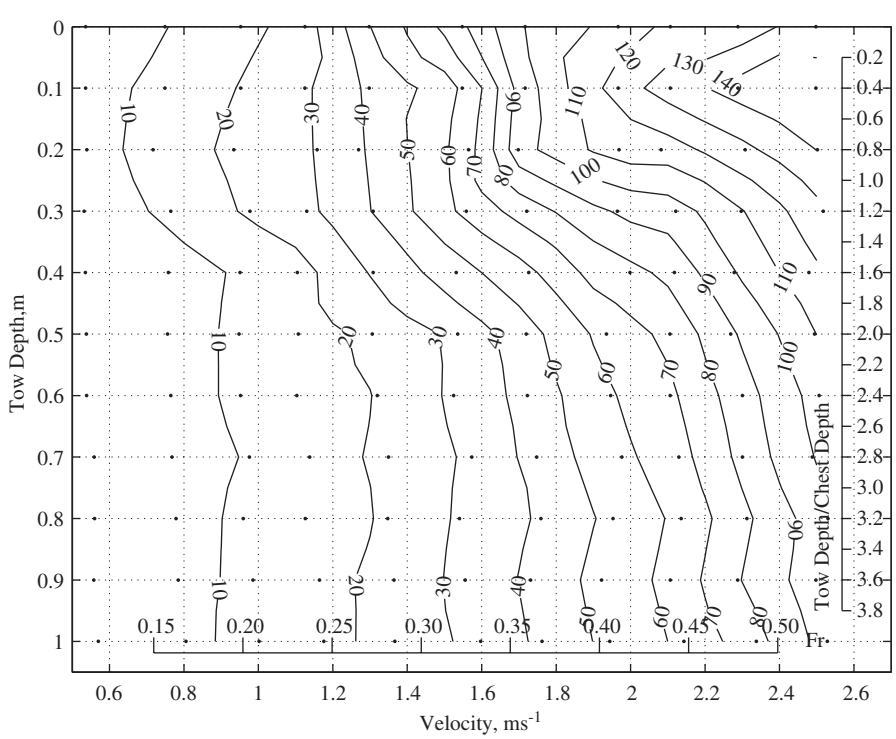

Fig. 5. Contours of passive drag for male mannequin as a function of velocity and depth. Dots indicate individual tows. The Froude number axis is based on finger to toe length. The right-hand axis gives normalized depth based on mannequin's chest depth of 0.25 m.

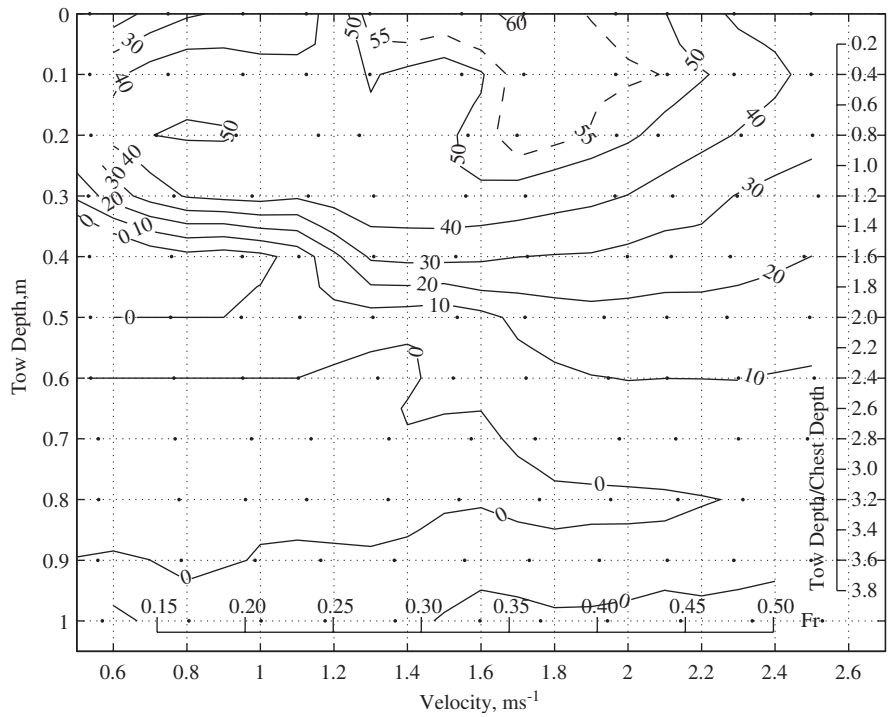

Fig. 6. Contours of additional drag near the surface as a function of tow depth and velocity, i.e. the percentage of drag due to waves, calculated as the total drag minus fully immersed drag at the same velocity, as a percentage of total drag. The Froude number axis is based on finger to toe length. The right-hand axis gives normalized depth based on mannequin's chest depth of 0.25 m.

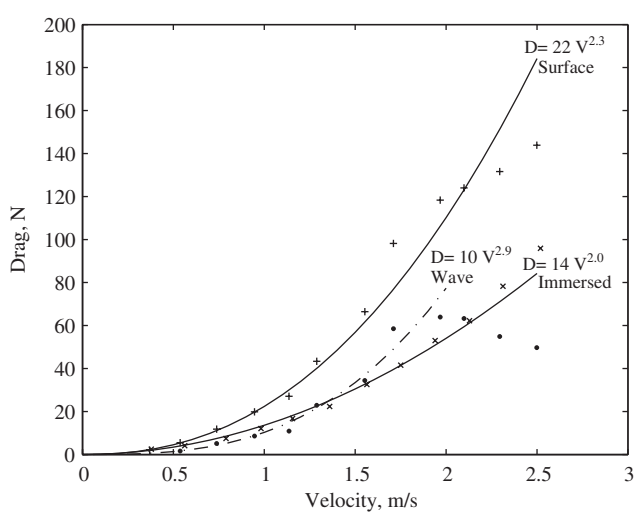

Fig. 7. Fit of passive drag as a power function of velocity, "+" and solid curve total drag averaged over 0.2–0.0 m tows. "×" and solid curve fully immersed drag curve; average of 1.0–0.8 m tows. Dots and chain dashed line are for the difference between total and immersed drag, i.e. wave drag. Equations give power law fits to measurements. For wave drag only data at velocities less than $2.0\,\mathrm{m\,s^{-1}}$ were fitted.

where a and n are constants. Fig. 7 shows this relationship fitted to the measurements using linear least squares. The uppermost solid curve is the total drag averaged over tow depths 0.2–0.0 m, where $n = 2.3$ and $a = 22$. The downward curvature of the drag curves above $2\,\mathrm{m\,s^{-1}}$ distorts the fit. However, fitting only the measurements below $2\,\mathrm{m\,s^{-1}}$ gives very similar results with $n = 2.4$ and $a = 23$. The lower solid curve is the fit to the fully immersed drag curve which increases more gradually with $n = 2.0$ and $a = 14$, thus obeys a V^2 power law. The difference between the surface and fully immersed drag is shown by the solid dots and a dashed fitted curve. This difference decreases at high speed as surface drag is smaller relative to immersed drag, thus only the differences for tows less than $2\,\mathrm{m\,s^{-1}}$ were fitted. For velocities up to this speed the additional drag experienced at the surface increases rapidly with velocity, $n = 2.9$ and $a = 10$.

4. Discussion

The drag curves and contour plots generally show increasing drag as the mannequin approaches the surface and it is important to determine the cause of this additional drag. Research on vessels shows that wave drag becomes increasingly more important with speed (Van Manen and Van Oossanen, 1988), also, wave drag on an immersed body increases as it approaches the surface. Thus the low-speed drag curves for all tow depths should coincide and at higher speeds should fan out above the fully immersed curves, with a steeper gradient at shallower tow depths. The mannequin's

measured drag curves generally show this pattern and thus are indicative of the influence of wave drag.

The bumps and hollows in the idealized drag curve in Fig. 1 are seen in model tests of idealized long parallel-sided flat-bottomed vessels and sharp ends (Newman, 1977). The bumps result from interference of the single bow with the model's single stern wave. Measured drag curves on wide model vessels with continuously varying width and draft often do not show clear bumps and hollows, as every change in cross-section creates an additional wave, smearing out the bumps and hollows due to the bow and stern wave interference. It is unlikely that the complex shape of the human body would create clear bumps and hollows in its drag curve.

The overall character of the curves for the shallowest three tows in Fig 3 is of a single bump and resembles the typical drag curve for a surface vessel as shown in Fig. 1, as they exhibit high upward curvature at lower velocities, $Fr = 0.20$–0.40 (0.9–$1.9\,\mathrm{m\,s^{-1}}$), and a slower increase or downward curvature for $Fr > 0.42$ ($>2.0\,\mathrm{m\,s^{-1}}$). Thus at the shallower depths and highest speeds, $Fr > 0.45$ ($>2.2\,\mathrm{m\,s^{-1}}$), the mannequin may be generating some dynamic lift as it moves towards planing, resulting in a more gradual increase in drag with velocity at high speed. Consequently, the character of the shallow drag curves is consistent with the higher drag being due to waves as the mannequin approaches planing. The drag curves suggest that if the mannequin's hands were free to move vertically it would fully plane on the surface at speeds above $2.6\,\mathrm{m\,s^{-1}}$. However, this is far faster than the speeds attainable by self-propelled swimmers.

Further support for identifying the additional drag as being due to waves is provided by theoretical estimates of the flow disturbance by the immersed mannequin. The frictionless flow around an ellipsoid was calculated using the method in Kundu and Cohen (2002). For an ellipsoid with the same frontal area and volume as the mannequin, maximum velocities were 7% higher at its widest point than in the undisturbed flow. At 2.0 diameters from the ellipsoid's axis the disturbance is reduced to less than 1% of the undisturbed velocity. Thus the ellipsoid should start to experience wave drag when it comes within 2.0 diameters of the water's surface. In Fig. 6 the 10% contour lies around 2.0 chest depths, supporting wave drag as the cause of the increased drag.

Examining the ellipsoid analogy more closely using Bernoulli, the 1% velocity disturbance due to an ellipsoid at 2.0 diameters depth would give a 0.006 m surface depression at $1.2\,\mathrm{m\,s^{-1}}$. To get the same depression at $2.2\,\mathrm{m\,s^{-1}}$, the ellipsoid would have to be at 2.38 diameters depth. For the mannequin this means wave drag at $1.2\,\mathrm{m\,s^{-1}}$ and 0.5 m depth should be the same as the wave drag at $2.2\,\mathrm{m\,s^{-1}}$ and 0.59 m. Fig. 6 shows that the 10% contour is near 0.5 m at $1.2\,\mathrm{m\,s^{-1}}$ and passes through 0.6 m at $2.2\,\mathrm{m\,s^{-1}}$, thus the slope of the 10% contour is consistent with wave drag. The ellipsoid analogy would only be expected to hold when the flow disturbance at the surface is weak, thus the differing slopes of the higher contours in Fig 6 may reflect the non-linear interaction of the disturbed flow with the surface.

The extent to which the observed changes in drag with tow depth can be attributed to wave drag rather than the other two main drag components can also be examined. For tow depths less than 0.4 m at lower speeds, parts of the mannequin are above the water, which may affect both skin and form drag. The areas above water were up to 25% of the total surface area, and might be expected to result in a reduction in skin drag, thus the observed increased drag is unlikely to be due to skin drag. As the mannequin nears the surface form drag is likely to change, whether this is an increase or decrease is not clear. However, the change in flow structure is likely to be small as the cross-sectional area exposed to the flow at the surface would be about the same or slightly smaller than when fully immersed. Consequently, the increased drag observed is most likely due to waves, but new measurement techniques would be needed to definitively isolate the wave component of drag. A similar difficulty in isolating drag components occurs in model testing of ships, where wave drag is usually isolated by calculations based on Michell's (1898) theory for slender ships (Tuck, 1989), a technique verified by measurements of the wave pattern behind models (see Van Manen and Van Oossanen, 1988).

Wilson and Thorp (2003) associated wave drag with the V^4 component of the drag–velocity curve to estimate wave drag as 40% of the total drag at $2.0\,\mathrm{m\,s^{-1}}$. The power law curve for the difference between surface and fully immersed drags (i.e. the wave drag in Fig. 7) grows at $V^{2.9}$ below $2\,\mathrm{m\,s^{-1}}$. Thus Wilson and Thorp's association does not appear to be valid, accounting for their under-estimating wave drag. Experiments with slender idealized wedge-shaped models with parallel mid-sections, show that wave drag grows as V^6 plus oscillatory terms due to wave interference (see Van Manen and Van Oossanen, 1988). Our measurements on a non-slender realistic body shape, with abrupt changes in cross-section, exhibits a V^3 relationship for wave drag speeds below $2.0\,\mathrm{m\,s^{-1}}$ and a more complex relationship at high speeds where both the total and wave drag curves have downward curvature in Fig. 7. Toussaint et al.'s (2002) estimate of 21% wave drag at $1.9\,\mathrm{m\,s^{-1}}$ on active swimmers was based on the assumption of zero wave drag at $1.6\,\mathrm{m\,s^{-1}}$ and a V^2 power law fit to the measurements below this velocity. They associated the difference between the extrapolation of the fit and measured total drag at $1.9\,\mathrm{m\,s^{-1}}$ with wave drag. Our measurements, however, show that passive wave drag at $1.6\,\mathrm{m\,s^{-1}}$ at the surface is more than 50% of the total.

Thus it is likely that wave drag on active swimmers at 1.6 m/s is much greater than zero and consequently, Toussaint et al.'s (2002) estimate significantly underestimates wave drag on active swimmers. Lyttle et al. (1998) showed a 20% increased in passive drag at the surface compared to a depth of 0.6 m at $2.2\,\mathrm{m\,s^{-1}}$, whereas our measurements show 10% wave drag at 0.6 m and wave drag as 50% of the total at the surface at $2.2\,\mathrm{m\,s^{-1}}$.

The drag curves in Fig. 3 at low speeds, $<1.2\,\mathrm{m\,s^{-1}}$, show two distinct groups rather than a progressive increase in drag due to waves as the surface is approached. The two groups are those at and above 0.3 m and those below 0.3 m. Observations showed that below 0.3 m the mannequin's hips were immersed at all speeds and at 0.3 m and above, the hips were above the surface at low speeds, $<1.2\,\mathrm{m\,s^{-1}}$. Thus drag is higher in the upper group where the mannequin is partially above water than when it is fully covered by water in the lower group. Within the near surface group the 0.0 m tow has the lowest drag and the remaining have similar drags. This may be a secondary effect of the hips being at the surface combined with a deeper tow depth, giving a slightly oblique towing angle which increases drag for tow depths 0.3–0.1 m. A consequence of the separation into two groups is that drags near the surface at low speeds appear to be twice those when fully immersed. Wave drag should be relatively small at low speeds and surface drags would be expected to be close to the immersed drags. Thus the additional drag when near the surface below $1.2\,\mathrm{m\,s^{-1}}$ in Fig. 6 may be due to oblique towing angles or other surface effects and not to wave drag. Further experiments are planned to confirm this.

The measurements presented here show high wave drag on passive/towed swimmers and it is important to consider how this relates to active/self-propelled swimmers. In active swimming the vertical components of the thrust forces may reduce the average displacement of the upper torso, reducing the body's bow wave and hence wave drag. However, rolling of the body and arm action of the active swimmer will generate a new set of waves. This new set is an additional wave drag not present in passive swimming. Whether there is a net reduction or gain of wave drag on active swimmers due to these two effects, when compared to passive swimmers, is an open question. To determine the net effect a different technique to that used here would be required. However, the measurements presented here strongly indicate that wave drag is also likely to be a substantial part of total drag on active swimmers.

5. Conclusions

The measurements show that total drag increases rapidly as the mannequin is towed at depths shallower than 2.8 chest depths (~0.7 m), at some velocities reaching up to 2.4 times the drag when the mannequin is fully immersed. Near or at the surface the reduced wetted surface and a similar form wake size suggest that the large increase in drag near the surface is unlikely to be due to skin or form drag. The upward and then downward curvature above $2.0\,\mathrm{m\,s^{-1}}$ ($Fr = 0.42$) of the near surface drag curves is similar to that of a high-speed surface vessel, for which wave drag dominates. In addition, the increase in drag starts at a depth at which the disturbed flow around the mannequin should begin to impinge on the surface and the slope of the 10% contour in Fig. 6 is consistent with wave drag. Thus we interpret the increased drag as the mannequin was towed at shallow depths as most likely due to wave drag and thus wave drag is the largest component of drag on surface swimmers at speeds above $Fr = 0.3$ ($\sim 1.4\,\mathrm{m\,s^{-1}}$). The mannequin was towed face up, thus the results are most applicable to the glide in backstroke swimming, however, wave drag is mainly influenced by the immersed volume which is the same face up or down when fully immersed and likely very similar at the surface. Thus high wave drag percentages are likely similar when towed face down.

The characteristics of the measured drag when the mannequin is towed near the surface are complicated, probably due to its progressive immersion from the head down as speed increases. How the measured drag applies to a flexible human swimmer whose posture may be affected by the fluid forces should be interpreted with care. The mannequin allows isolation of the drag differences due to depth and velocity from those due to differences in streamlined posture which are of order 10%, much smaller than the differences due to depth. The value of the comprehensive measurements presented here lies in demonstrating the large difference between surface and immersed drag rather than in predicting absolute drag on real swimmers.

The principle finding is that wave drag is 50–60% of the total passive drag force on elite swimmers at the surface, much higher than any previous estimate. This is likely to be similar for active swimmers and thus the development of new techniques to reduce wave drag could have significant impact on performance. In addition, it is shown that swimmers must be more than 1.8 chest depths below the surface at $Fr = 0.2$ ($0.9\,\mathrm{m\,s^{-1}}$) and 2.8 chest depths at $Fr = 0.42$ ($2.0\,\mathrm{m\,s^{-1}}$), during underwater sections of starts and turns to avoid significant wave drag.

Acknowledgments

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